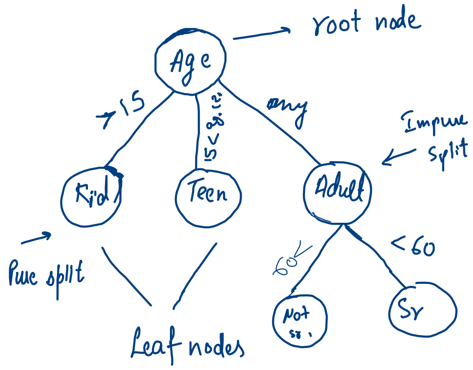
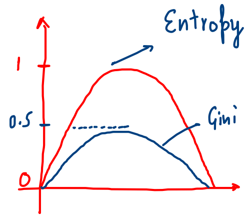
Decision Trees

Decision Trees

```
if age < 15 :
    print ("He is a kid")
elif 15 < age < 20 :
    print ("Teen")
else :
    print ("Adult")
```



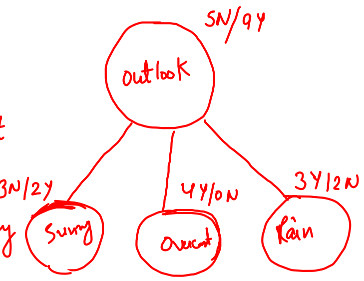


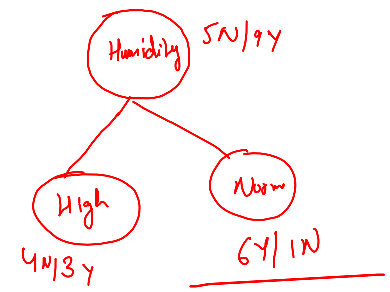
→ Target / Dependent

① → Pure / Impure split
- Entropy
- Gini Impurity

② feature selection
- Information Gain

$= 0.6 \times 0.73 + 0.4 \times 1.3 = 0.95 \rightarrow 0.94$





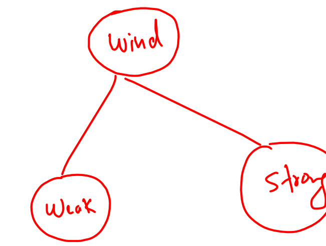


Table 1. Play Tennis dataset

Outlook	Humidity	Wind	Play
Sunny	High	Weak	No
Sunny	High	Strong	No
Overcast	High	Weak	Yes
Rain	High	Weak	Yes
Rain	Normal	Weak	Yes
Rain	Normal	Strong	No
Overcast	Normal	Strong	Yes
Sunny	High	Weak	No
Sunny	Normal	Weak	Yes
Rain	Normal	Weak	Yes
Sunny	Normal	Strong	Yes
Overcast	High	Strong	Yes
Overcast	Normal	Weak	Yes
Rain	High	Strong	No

1) ⓐ Entropy $H(s) = -p_0 \log p_0 - p_1 \log p_1$

$$\boxed{H(s) = -\sum_{i=0}^{n} p_i \log p_i}$$ Range → [0, 1]

ⓑ $$\boxed{\text{Gini Impurity} = 1 - \sum p_i^2}$$ Range → [0, 0.5]

$H(\text{Sunny}) = -\frac{3}{5}\log_2\left(\frac{3}{5}\right) - \frac{2}{5}\log_2\left(\frac{2}{5}\right) = 0.94$ (Impure Split)

$H(\text{Overcast}) = -\frac{0}{4}\log_2\left(\frac{0}{4}\right) - \frac{4}{4}\log_2\left(\frac{4}{4}\right) = 0$ (Pure split)

$H(\text{Rain}) \simeq H(\text{Sunny})$

$\text{Gini}(\text{Sunny}) = 1 - \left[\left(\frac{2}{5}\right)^2 + \left(\frac{3}{5}\right)^2\right]$

$= 1 - [0.16 + 0.36]$

$= 1 - [0.52] = 0.48$

$\text{Gini}(\text{Overcast}) = 1 - \left[\left(\frac{4}{4}\right)^2 + \left(\frac{0}{4}\right)^2\right]$

$= 1 - [1 + 0] = 0$

② Information Gain → Means which feature should we start with

$$\text{Information Gain} = H(\text{root}) - \sum_{i=1}^{n} \frac{N_i}{N} H(\text{child}_i)$$

$$\text{Info Gain}(\text{Outlook}) = -p_y \log_2 p_y - p_n \log_2 p_n - \sum_{i=1}^{K}\left[\frac{\text{Count of Samples in Child}}{\text{Total number of Samples in root}} \times \text{Entropy of Child}\right]$$

$$= -\frac{9}{14}\log_2\frac{9}{14} - \frac{5}{14}\log_2\frac{5}{14} - \left[\frac{5}{14} \times H(\text{Sunny}) + \frac{4}{14} \times H(\text{overcast}) + \frac{5}{14} \times H(\text{Rain})\right]$$

$$\Rightarrow 0.64 \times 0.63 + 0.35 \times 1.48 - [0.35 \times 0.94 + 0.28 \times 0 + 0.35 \times 0.94]$$

$$\Rightarrow 0.40 + 0.51 - [0.329 + 0 + 0.329] = 0.91 - 0.658$$

$$\Rightarrow 0.252$$

$$\text{Info Gain}(\text{Humidity}) = -H(\text{Humidity}) - \sum_{i=1}^{n} \frac{\text{Number of Child}_i\text{ Samples}}{\text{Total no. of Humidity Samples}} \times H(\text{Child}_i)$$

$$\simeq 0.3$$

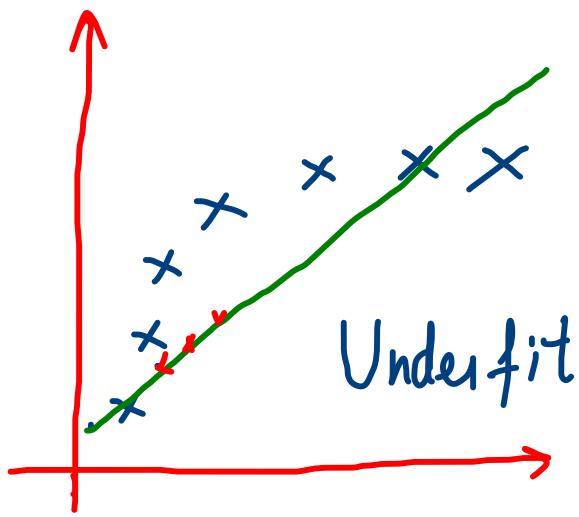
Overfitting vs Underfitting - Bias Variance Tradeoff

Bias: Training data

Variance: Test data

Regression

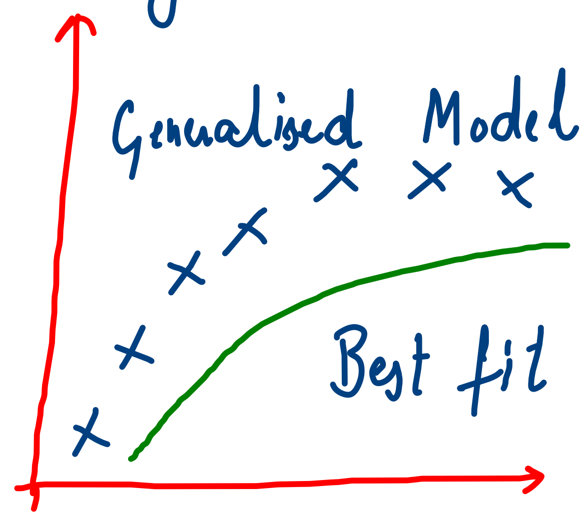
Classification



Training Error ↑↑

High Bias
+
Low Variance

High Variance

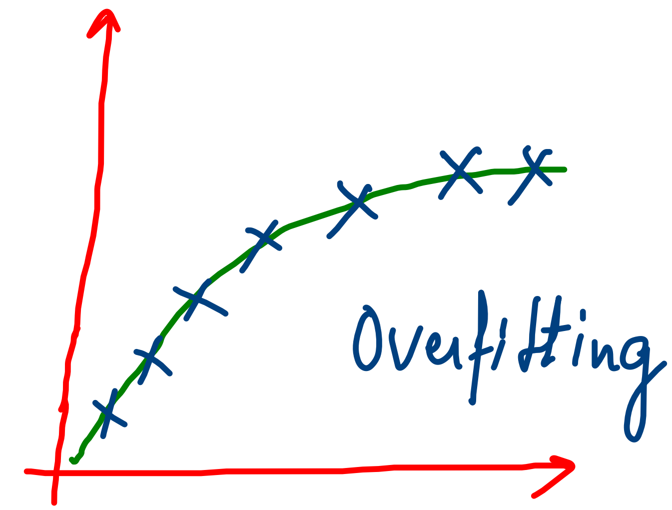


Training Error ↓↓

Low Bias

Test Error ↓↓

Low Variance



Training Error ↓↓ ≈ 0

Low Bias

Test Error ↑↑

High Variance